

Product Requirements Document Ride Connect

1. Introduction

This document outlines the requirements for RideConnect, a new ride-sharing application designed to connect riders with drivers efficiently and safely.

2. Product Overview

RideConnect will offer a mobile application for both riders and drivers, facilitating ride requests, driver-rider matching, it could be used for any purpose, but we assume it will be Uber-Like experience, and will not be used for order delivery (matching between driver and customer order) or any other scenario. We will only focus on ride requests and acceptance, disregarding payments, ratings, providing optimal routes, and sophisticated estimations and recalculations of fare

3. Functional Requirements

3.1 Driver & Rider Management

- Collect and store necessary user information:
 - For riders: Name, contact details
 - For drivers: Name, contact details, vehicle information (make, model, color, plate number), driver's license information
- Implement driver status management (online, busy, offline, suspended)
- Driver is busy once he accepts a ride

3.2 Location and Mapping

- Capture and maintain up-to-date location information for riders and drivers
- Enable riders to input pickup and drop-off locations for ride requests
- Provide **nearby** available drivers to riders within 5 km radius of rider pickup
- Show estimated time of arrival for drivers to pickup locations based on the distance between the pickup and drop-off

3.3 Pricing

- Calculate ride fare based on base rate of the vehicle (economy, premium, family), distance, time, and any applicable surcharges
- Economy adds 0% to the total fare, premium 10% and family 25%
- If the pick up is in a surge area, the total fare will increase based on the surge area multiplier
- Consider the distance in the pricing by assuming the speed is 40 km/h

3.4 Ride Request and Matching

- Allow riders to submit ride requests specifying pickup and drop-off locations
- Update the ride with the estimated arrival time based on the distance and by assuming the speed is 40 km/h, and update the estimated duration
- Update the ride with the estimated price
- Identify suitable drivers for each ride request based on **proximity** and **availability**
- Notify selected drivers of new ride requests
- Drivers would then accept or decline ride requests
- Match riders with the first driver to accept their request
- Update the ride with matched driver

3.5 Ride Management

- Track the status of each ride (*requested, accepted, in progress, completed, cancelled*)
- Record key ride information:
 - Pickup and drop-off locations
 - Start and end times
 - Distance traveled
 - Route taken for traceability purposes
 - Ride duration